Lecture 11: Multiple Linear Regression

PSTAT100: Data Science — Concepts and Analysis

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🚁 Overview

Aims of the lecture

- Introduce **multiple linear regression (MLR)**: modelling Y as a linear function of multiple predictors $X_1, \ldots, X_p$.
- Derive the **OLS estimator** $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.
- Establish **properties** of MLR OLS:
  - Unbiasedness
  - Variance-covariance matrix, and BLUE.
- Conduct inference with t**-tests** and the **overall** F **-test**.
- Measure fit with **adjusted** R^2 and information criteria (AIC, BIC).

📚 Required Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f as f_dist, t as t_dist
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.datasets import fetch_california_housing
```

💅 Figure Styles

```
sns.set_style('whitegrid')
sns.set_palette('Set2')
```

Multiple Linear Regression

🔁 Why Multiple Predictors?

Maximize use of information

- Most real phenomena are driven by **multiple factors** simultaneously.
- We wish to capture the **multivariate relationships**.
- Omitting relevant predictors leads to the SLR slope absorbing the effect of missing variables that are correlated with X.
    - This is known as **omitted variable bias**.

Recall from Lecture 10

> The SLR residual plot for body mass vs. flipper length showed **systematic banding** — a strong hint that **species** was acting as an omitted variable.

- Adding species and other measurements can:
    - Reduce bias in coefficient estimates.
    - Increase the proportion of variance explained.
    - Improve prediction accuracy.

The MLR Model

Multiple Linear Regression Model

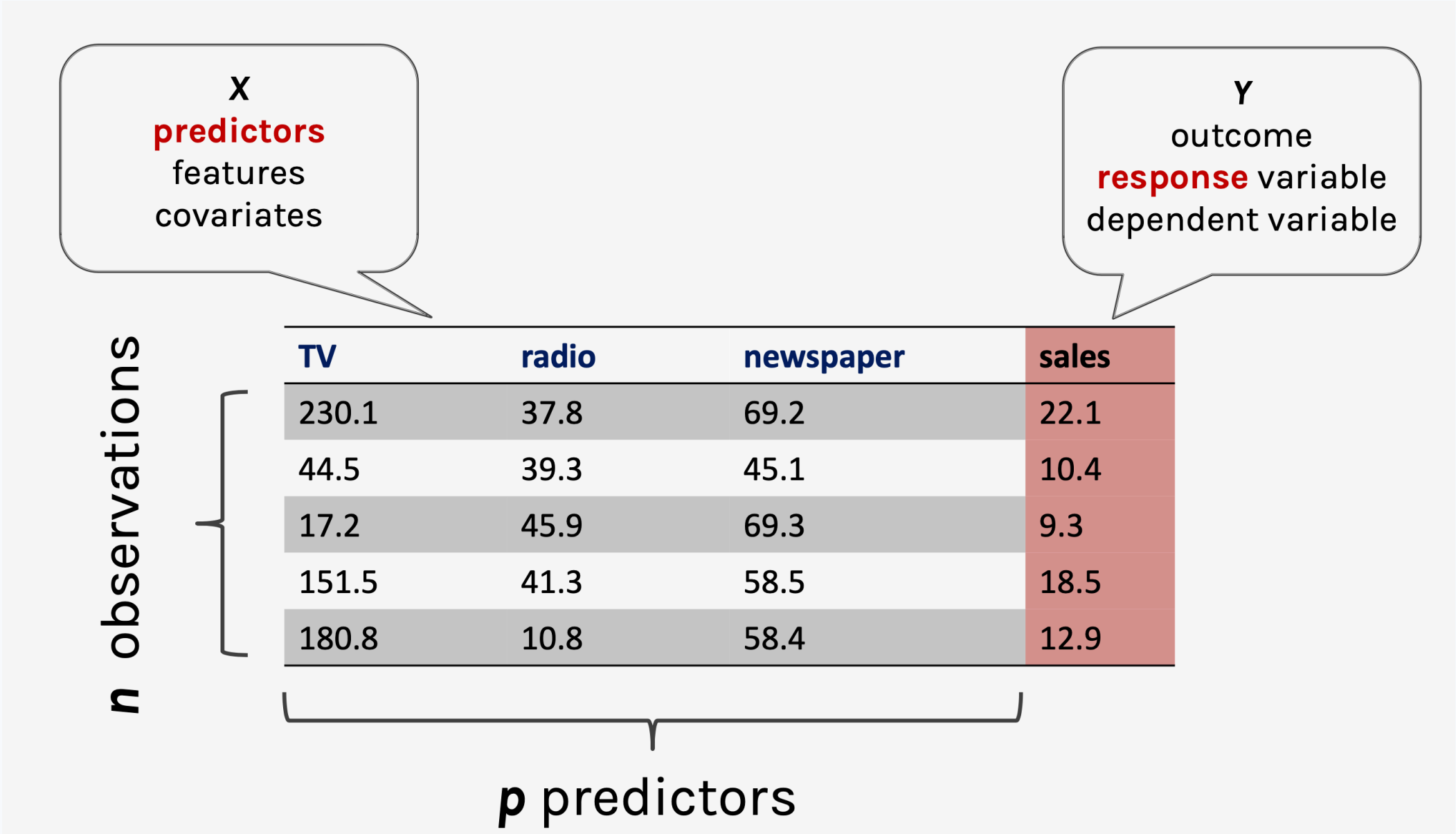
For response y with p predictors $x_1, \ldots, x_p$, the MLR model is defined as

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where $\varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$ (under the Gauss-Markov assumptions).

- There are $p + 1$ parameters: one **intercept** β_0 and p **slopes** $\beta_1, \ldots, \beta_p$.
- There are n observations: $\{(y_i, x_{i1}, \ldots, x_{ip})\}_{i=1}^n$.
- SLR is the special case $p = 1$.
- We require $n > p + 1$ observations to fit the model.

MLR Table



TV	radio	newspaper	sales
230.1	37.8	69.2	22.1
44.5	39.3	45.1	10.4
17.2	45.9	69.3	9.3
151.5	41.3	58.5	18.5
180.8	10.8	58.4	12.9

Example of observations, predictors and response for MLR.

Partial Slopes — A Critical Difference

SLR vs MLR

- In SLR, $\hat{\beta}_1$ captures the total marginal relationship between X and Y.
- In MLR, each slope is a **partial slope**:
  - β_j is the expected change in Y for a **one-unit increase in X_j, holding all other predictors constant** (ceteris paribus).

Example

- In SLR of body mass on flipper length, $\hat{\beta}_1$ includes the indirect effect of species (Gentoo penguins are larger *and* have longer flippers).
- In MLR with both flipper length and species, $\hat{\beta}_{\text{flipper}}$ measures the flipper–mass relationship **within the same species**.
- This is why partial slopes can differ substantially — even in sign — from SLR slopes.

Matrix Notation

To simplify notation, we will express the MLR model in **matrix form**. We define the following:

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}. \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

Then we can write our model as

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

where we have that $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

- Note here that $\boldsymbol{X}$ has $p + 1$ columns (including the intercept) and n rows (observations).
- This matrix formulation is more compact and allows us to derive the **OLS estimator** using linear algebra.

Gauss-Markov Assumptions

Gauss-Markov assumptions in matrix form

Assumption	Matrix statement
Linearity	$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$
Independence + Homoscedasticity	$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$
Normality	$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0},\, \sigma^2 \mathbf{I}_n)$
Full rank	$\mathrm{rank}(\mathbf{X}) = p + 1$ (columns linearly independent)

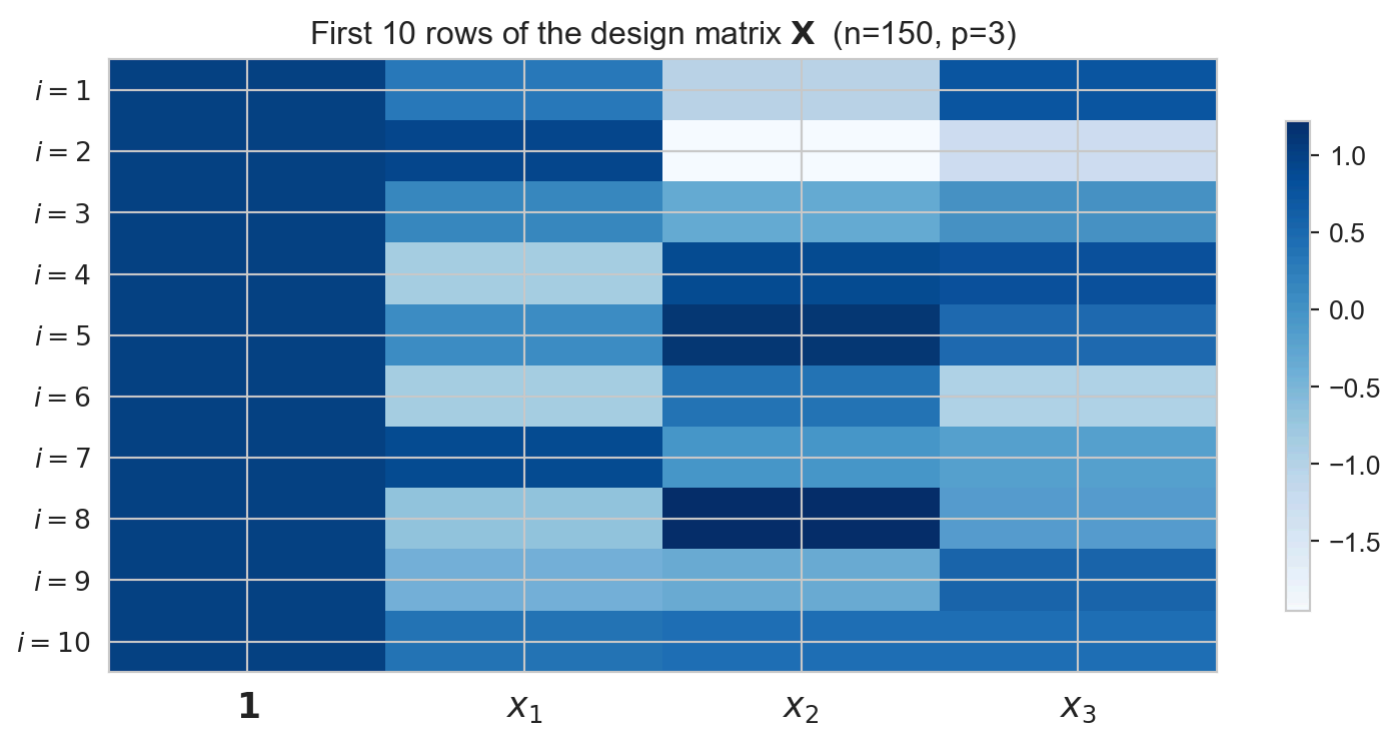
- Together these imply

$$\mathbf{Y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta},\, \sigma^2 \mathbf{I}_n).$$

Constructing the Design Matrix in Python

```
# Set seed and define true parameters
rng = np.random.default_rng(42)
n, p = 150, 3
beta0_true = 3.0 # intercept
beta_true = np.array([2.0, -1.5, 0.8]) # slopes
sigma_true = 2.0 # noise level

# Define the design matrix and response vector
X_pred   = rng.normal(0, 1, size=(n, p))
X_design = np.column_stack([np.ones(n), X_pred])
y = X_design @ np.r_[beta0_true, beta_true] + rng.normal(0, sigma_true, size=n)

# Plot the first 10 rows of the design matrix
fig, ax = plt.subplots(figsize=(8, 4))
im = ax.imshow(X_design[:10], aspect='auto', cmap='Blues')
ax.set_xticks(range(p + 1))
ax.set_xticklabels(['$\\mathbf{1}$', '$x_1$', '$x_2$', '$x_3$'], fontsize=13)
ax.set_yticks(range(10))
ax.set_yticklabels([f'$i = {i+1}$' for i in range(10)])
ax.set_title('First 10 rows of the design matrix $\\mathbf{X}$  (n=150, p=3)')
plt.colorbar(im, ax=ax, shrink=0.8)
plt.tight_layout(); plt.show()
```

Constructing the Design Matrix in Python

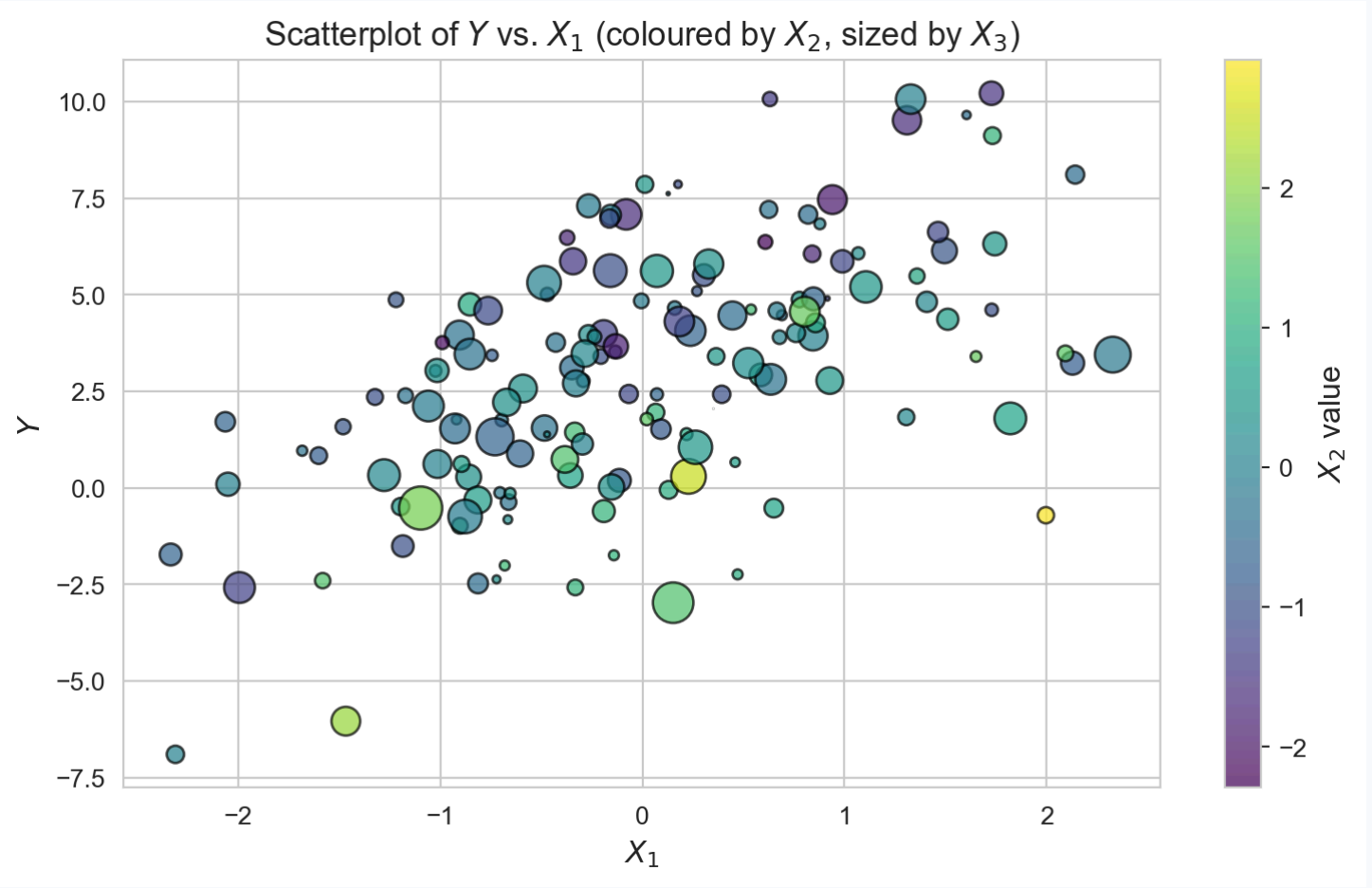


Heatmap of the first 10 rows of the design matrix X (p = 3 predictors). The first column is all ones (intercept term).

Scatterplot of Y vs. X_1 (with points coloured by X_2 and sized by X_3) to illustrate the multivariate relationship.

```
plt.figure(figsize=(8, 5))
scatter = plt.scatter(X_pred[:, 0], y, c=X_pred[:, 1], s=100 * np.abs(X_pred[:, 2]),
                      cmap='viridis', alpha=0.7, edgecolor='k')
plt.xlabel('$X_1$', fontsize=12)
plt.ylabel('$Y$', fontsize=12)
plt.title('Scatterplot of $Y$ vs. $X_1$ (coloured by $X_2$, sized by $X_3$)', fontsize=13)
cbar = plt.colorbar(scatter)
cbar.set_label('$X_2$ value', fontsize=12)
plt.tight_layout(); plt.show()
```

Scatterplot of Y vs. X_1 (with points coloured by X_2 and sized by X_3) to illustrate the multivariate relationship.



Scatterplot of Y vs. X_1, coloured by X_2 and sized by X_3. The relationship between Y and X_1 depends on the values of the other predictors, illustrating the need for MLR.

Pairplot of Y and the three predictors to visualise their relationships.

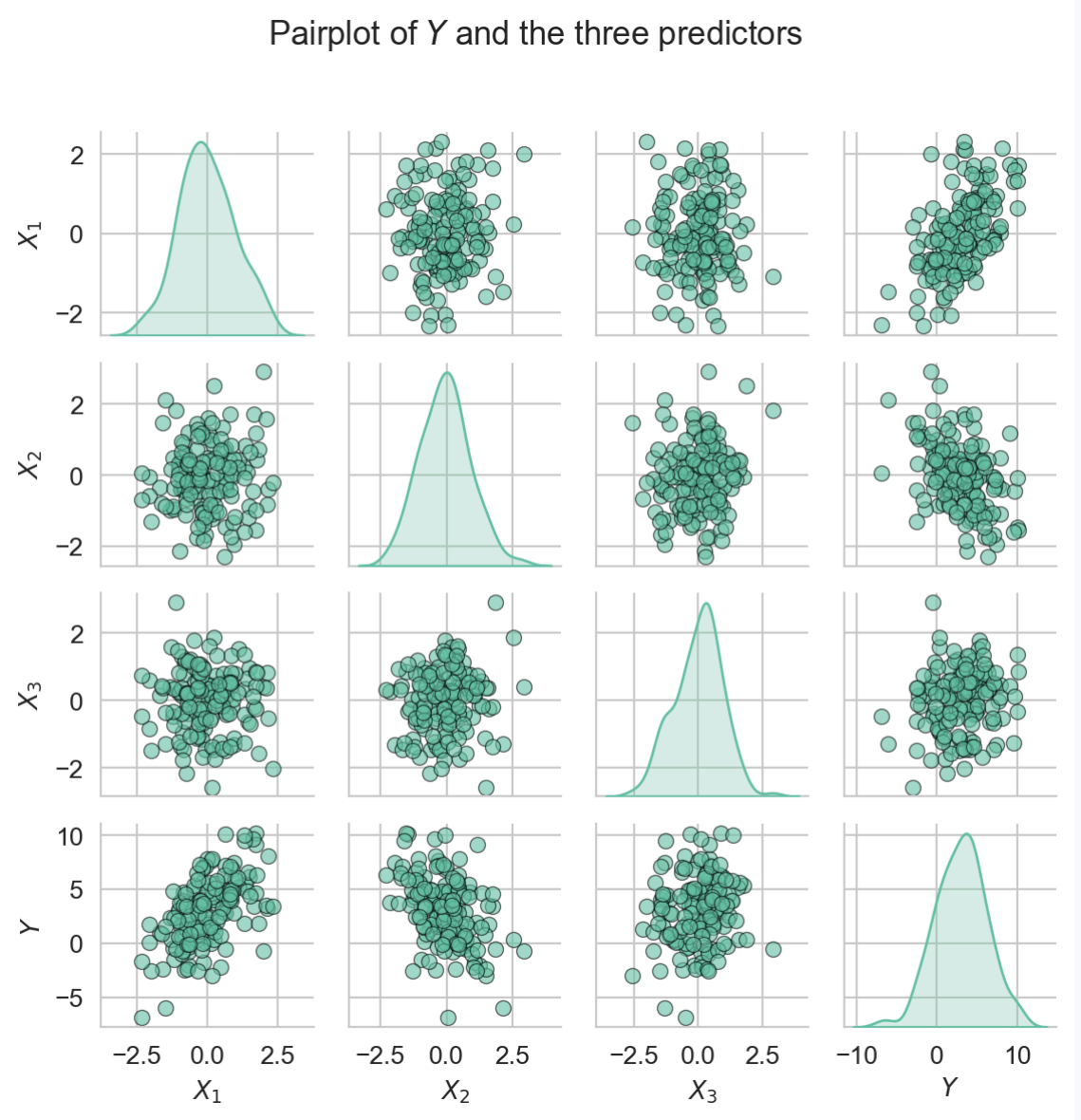
Correlation Matrix

```
df = pd.DataFrame(
    X_pred,
    columns=[f'$X_{j}$' for j in range(1, p + 1)]
    )
df['$Y$'] = y
print(df.corr().round(3))
```

```
       $X_1$  $X_2$  $X_3$    $Y$
$X_1$  1.000  0.055 -0.009  0.560
$X_2$  0.055  1.000  0.118 -0.432
$X_3$ -0.009  0.118  1.000  0.147
$Y$    0.560 -0.432  0.147  1.000
```

```
sns.pairplot(df, diag_kind='kde', plot_kws={'alpha': 0.6, 'edgecolor': 'k'}, height=1.5, aspect=1.0)
plt.suptitle('Pairplot of $Y$ and the three predictors', fontsize=13, y=1.02)
plt.tight_layout(); plt.show()
```

Pairplot of Y and the three predictors to visualise their relationships.



OLS Estimation in MLR

🎯 The OLS Criterion in Matrix Form

Once again we will estimate $\boldsymbol{\beta}$ using **ordinary least squares (OLS)**. The OLS estimator is the value of $\boldsymbol{\beta}$ that minimises the **residual sum of squares**:

$$\mathrm{RSS}(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

Expanding the objective we get that

$$\mathrm{RSS}(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Taking the **matrix gradient** with respect to β and setting it to zero gives the **normal equations**:

$$\mathbf{X}^\top \mathbf{X}\, \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

Provided $\mathbf{X}^\top \mathbf{X}$ is invertible (i.e. $\mathbf{X}$ has full column rank), the unique solution is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

> This derivation is beyond the scope of the course but we see again that our parameter estimates are given explicitly.

Connection to SLR ($p = 1$)

When $p = 1$ and $\mathbf{X} = [\mathbf{1},\ \mathbf{x}]$, we have:

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix}.$$

Computing $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ and simplifying (using $S_{xx} = \sum (x_i - \bar{x})^2$), we recover:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- MLR OLS is a **direct generalisation** of the SLR formulas from Lecture 9.

The Hat Matrix

The **fitted values** are:

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} = \mathbf{H}\mathbf{Y},$$

where $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$

- This is the **hat matrix** (it "puts a hat" on $\mathbf{Y}$).
- This matrix is central to understanding the **geometry** of OLS and the properties of residuals and is studied in depth in advanced linear regression courses.

Properties of $\mathbf{H}$

- **Symmetric**: $\mathbf{H}^\top = \mathbf{H}$.
- **Idempotent**: $\mathbf{H}^2 = \mathbf{H}$ — projecting twice is the same as projecting once.
- **Rank**: $\text{rank}(\mathbf{H}) = p + 1$.
- **Leverages**: diagonal entries $h_{ii} = [\mathbf{H}]_{ii}$ measure how unusual observation i's predictor values are. High leverage $\Rightarrow$ more influence on the fitted line.

Hat Matrix and Residuals

What is the Hat Matrix?

- The hat martix H is the orthogonal projection matrix onto the subspace spanned by the columns of $\boldsymbol{X}$.
  - The fitted values $\hat{\boldsymbol{y}}$ are the closest point in the column space of $\boldsymbol{X}$ to the observed $\boldsymbol{y}$.
  - This is definitely beyond the scope of the course but it is a fundamental geometric fact about OLS.

Hat Matric and Residuals

Since the residuals can be written as

$$\boldsymbol{e} = \boldsymbol{Y} - \hat{\boldsymbol{Y}} = (\boldsymbol{I} - \boldsymbol{H})\boldsymbol{Y}.$$

the variance of the residuals is

$$\mathrm{Var}(\boldsymbol{e}) = \sigma^2(\boldsymbol{I} - \boldsymbol{H}).$$

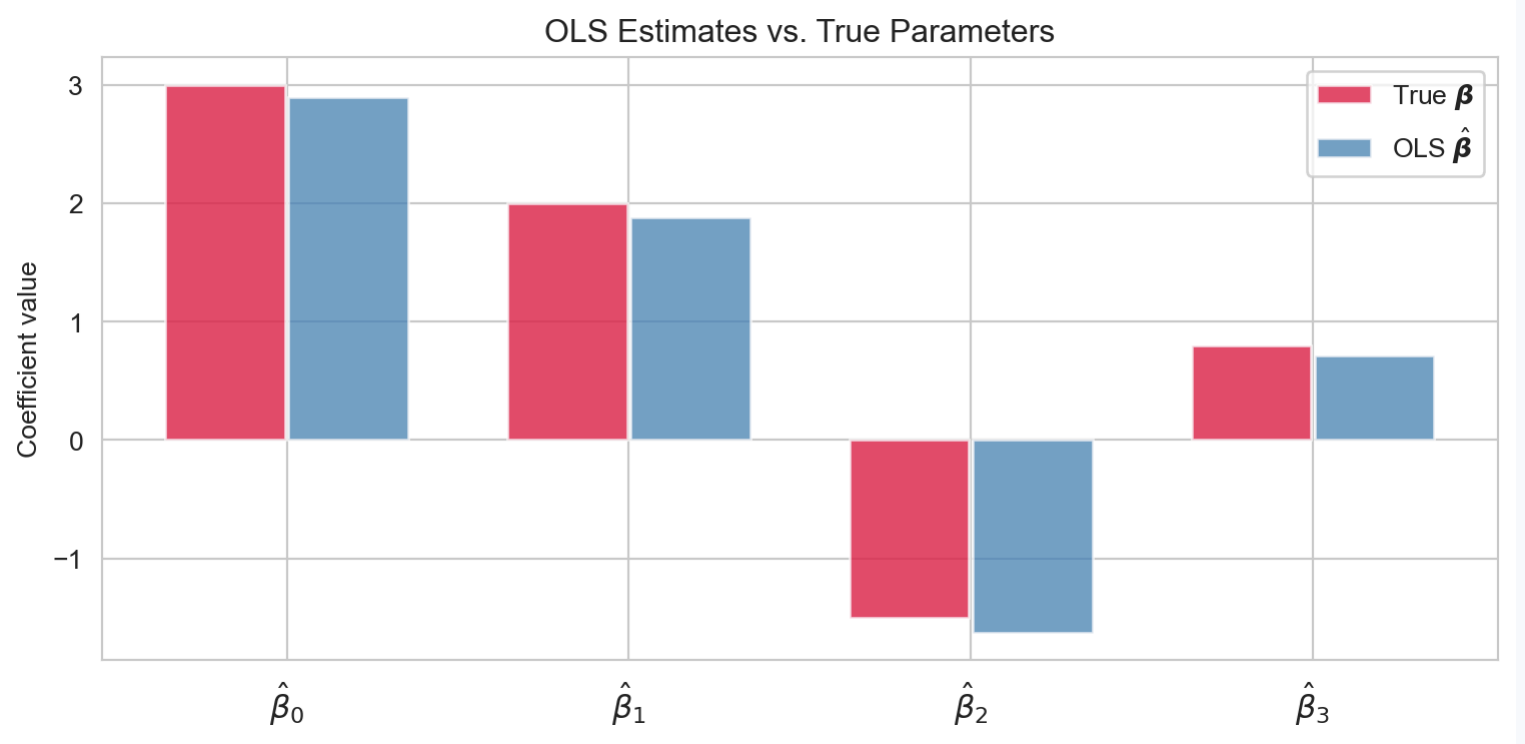
- Hence the residuals are not i.i.d. even when the errors are.

Computing OLS from Scratch

```python
# Solve the normal equations (numerically stable — avoids forming the explicit inverse)
beta_hat = np.linalg.solve(X_design.T @ X_design, X_design.T @ y)

# Compare to the true values
beta_all_true = np.r_[beta0_true, beta_true]
print("Parameter | True   | Estimated")
print("-" * 36)
for j, (tv, ev) in enumerate(zip(beta_all_true, beta_hat)):
    print(f"  β_{j}      |  {tv:5.2f}  |  {ev:8.3f}")

# Plot
labels = [f'$\\hat{{\\beta}}_{j}$' for j in range(p + 1)]
x_pos  = np.arange(p + 1)
fig, ax = plt.subplots(figsize=(8, 4))
ax.bar(x_pos - 0.18, beta_all_true, width=0.35,
       label='True $\\boldsymbol{\\beta}$', color='crimson', alpha=0.75)
ax.bar(x_pos + 0.18, beta_hat,      width=0.35,
       label='OLS $\\hat{\\boldsymbol{\\beta}}$', color='steelblue', alpha=0.75)
ax.set_xticks(x_pos); ax.set_xticklabels(labels, fontsize=12)
ax.set_ylabel('Coefficient value')
ax.set_title('OLS Estimates vs. True Parameters')
ax.legend(); plt.tight_layout(); plt.show()
```

Computing OLS from Scratch

```
Parameter | True   | Estimated
-----------------------------------
  β_0     |   3.00 |     2.896
  β_1     |   2.00 |     1.881
  β_2     |  -1.50 |    -1.626
  β_3     |   0.80 |     0.718
```



OLS estimates vs. true parameter values (p = 3). The estimator closely recovers all four true parameters.

Properties of the MLR OLS Estimator

✅ Unbiasedness

Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into the OLS formula:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \boldsymbol{\varepsilon}.$$

Taking expectations (using $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$ and treating $\mathbf{X}$ as fixed):

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}. \quad \checkmark$$

- Unbiasedness holds under GM1–GM3 (**normality is not required**).
- On average across many datasets, OLS returns the true parameter vector.

Variance-Covariance Matrix of $\hat{\beta}$

Collinearity

Collinearity refers to the case in which two or more predictors are correlated (related).

- When predictors are highly correlated, $\mathbf{X}^\top \mathbf{X}$ becomes close to singular, causing the OLS estimator to become unstable and have large variance.

Covariance of β

We wish to compute $\mathrm{Cov}(\hat{\boldsymbol{\beta}})$. We have that

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \boldsymbol{C}\boldsymbol{Y}.$$

Applying the linear transformation rule for variance we have

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = C\mathrm{Var}(\sigma^2 \boldsymbol{I})C^\top = \sigma^2 CC^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

What drives precision?

Properties of Variance-Covariance Matrix

- **Diagonal** entry j: $\mathrm{Var}(\hat{\beta}_j) = \sigma^2[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}$.
- **Off-diagonal** entries: covariances between different coefficient estimates.
- We estimate this matrix by substituting $\hat{\sigma}^2 = \mathrm{RSS}/(n-p-1)$.
- Standard errors: $\widehat{\mathrm{SE}}(\hat{\beta}_j) = \hat{\sigma}\sqrt{[(\mathbf{X}^\top\mathbf{X})^{-1}]_{jj}}$.

Precision Drivers

Factor	Effect on $\mathrm{Var}(\hat{\boldsymbol{\beta}})$
More observations ($\uparrow n$)	Smaller — more information
Less noise ($\downarrow \sigma^2$)	Smaller — cleaner signal
More spread in predictors	Smaller — better leverage
High multicollinearity	Larger — $(\mathbf{X}^\top\mathbf{X})^{-1}$ blows up

Gauss-Markov Theorem in MLR

Gauss-Markov Theorem (Matrix Form)

Under GM1–GM3 (**normality is not required**), the OLS estimator $\hat{\boldsymbol{\beta}}$ is **BLUE**:

For any other linear unbiased estimator $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{Y}$,

$$\mathrm{Var}(\tilde{\beta}_j) \geq \mathrm{Var}(\hat{\beta}_j) \quad \text{for all } j.$$

Estimating σ^2

Fitting $p+1$ parameters costs $p+1$ degrees of freedom. The **unbiased** estimator is:

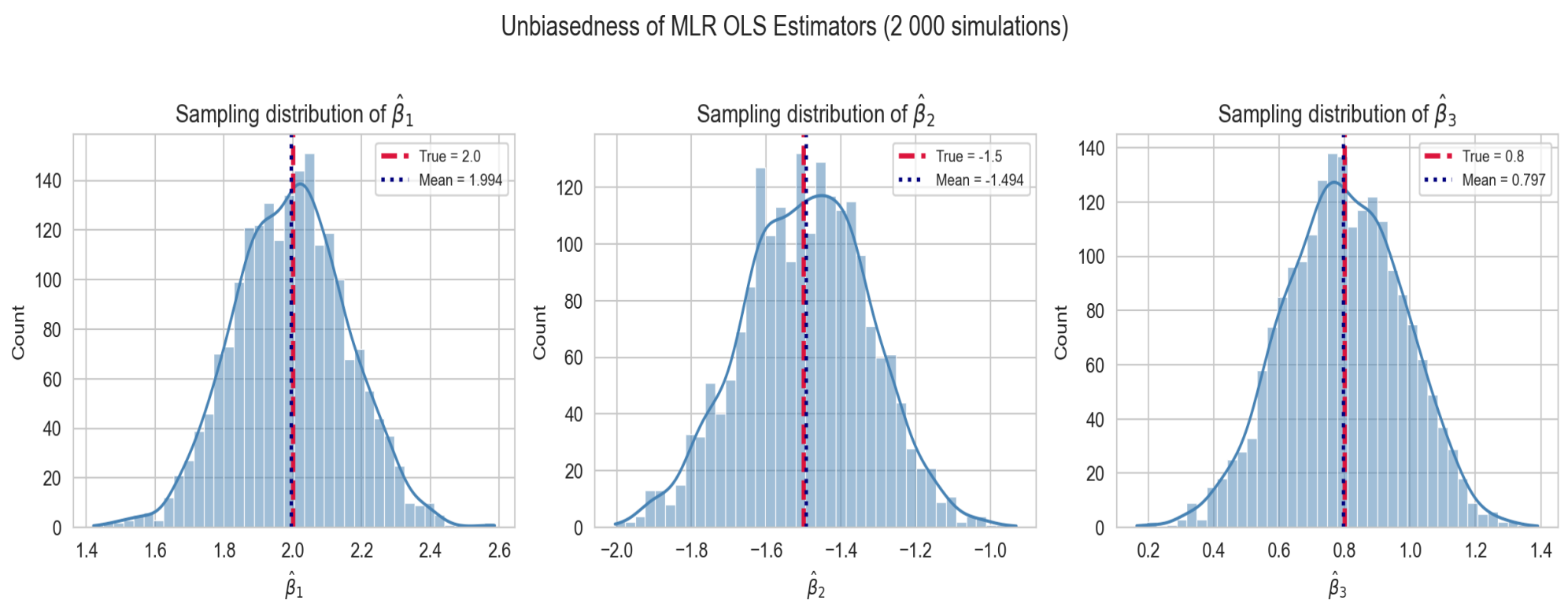
$$\hat{\sigma}^2 = \frac{\mathrm{RSS}}{n-p-1}.$$

We divide by $n-p-1$, not $n-2$ as in SLR.

Simulating the Sampling Distribution

```
rng2 = np.random.default_rng(7)
beta_hat_sims = []

for _ in range(2000):
    y_s  = X_design @ np.r_[beta0_true, beta_true] + rng2.normal(0, sigma_true, size=n)
    bh_s = np.linalg.solve(X_design.T @ X_design, X_design.T @ y_s)
    beta_hat_sims.append(bh_s)

beta_hat_sims = np.array(beta_hat_sims)   # shape (2000, 4)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
for j in range(3):
    lab = rf'$\hat{{\beta}}_{j+1}$'
    tv  = beta_true[j]
    sns.histplot(beta_hat_sims[:, j + 1], bins=40, kde=True,
                 color='steelblue', ax=axes[j])
    axes[j].axvline(tv, color='crimson', lw=2.5, linestyle='--',
                    label=f'True = {tv}')
    axes[j].axvline(beta_hat_sims[:, j+1].mean(), color='navy', lw=2, linestyle=':',
                    label=f'Mean = {beta_hat_sims[:, j+1].mean():.3f}')
    axes[j].set_xlabel(lab)
    axes[j].set_title(f'Sampling distribution of {lab}')
    axes[j].legend(fontsize=8)

plt.suptitle('Unbiasedness of MLR OLS Estimators (2 000 simulations)',
             fontsize=13, y=1.01)
plt.tight_layout(); plt.show()
```

Simulating the Sampling Distribution



Sampling distributions of $\hat{\beta}_1$, $\hat{\beta}_2$, $\hat{\beta}_3$ across 2 000 simulated datasets. Each estimator is centred on the true value (red dashed line), confirming unbiasedness.

Statistical Inference in MLR

📊 t-Tests for Individual Coefficients

Under GM1–GM4 (including normality):

$$T_j = \frac{\hat{\beta}_j}{\widehat{\mathrm{SE}}(\hat{\beta}_j)} \sim t_{n-p-1}.$$

The test and its interpretation

- Tests $H_0 : \beta_j = 0$ **given all other predictors are already in the model**.
- Equivalently: does X_j add explanatory power **beyond** what the other predictors already provide?
- A large p-value for $\hat{\beta}_j$ does **not** mean X_j is unrelated to Y — it may simply be collinear with the others.

```python
y_hat   = X_design @ beta_hat
RSS     = np.sum((y - y_hat)**2)
sigma_h = np.sqrt(RSS / (n - p - 1))
XtX_inv = np.linalg.inv(X_design.T @ X_design)
SE      = sigma_h * np.sqrt(np.diag(XtX_inv))

print(f"{'':5} {'β̂':>8} {'SE':>8} {'t':>7} {'p-val':>10}")
for j in range(p + 1):
    tv = beta_hat[j] / SE[j]
    pv = 2 * t_dist.sf(abs(tv), df=n - p - 1)
    print(f"β_{j}   {beta_hat[j]:>8.3f} {SE[j]:>8.4f}"
          f" {tv:>7.3f} {pv:>10.3e}")
```

```
            β̂       SE        t      p-val
β_0      2.896   0.1707  16.967  2.381e-36
β_1      1.881   0.1732  10.859  1.666e-20
β_2     -1.626   0.1813  -8.971  1.314e-15
β_3      0.718   0.1865   3.849  1.769e-04
```

The Overall F-Test

Tests whether **any** predictor is useful:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_j \neq 0.$$

Overall F-Statistic

$$F = \frac{\text{SSR}/p}{\text{SSE}/(n-p-1)} = \frac{R^2/p}{(1-R^2)/(n-p-1)} \sim F_{p,\,n-p-1} \quad \text{under } H_0.$$

```python
SST    = np.sum((y - y.mean())**2)
SSE    = np.sum((y - y_hat)**2)
SSR    = SST - SSE
R2     = SSR / SST

F_stat = (SSR / p) / (SSE / (n - p - 1))
p_val  = f_dist.sf(F_stat, dfn=p, dfd=n - p - 1)

print(f"R²      = {R2:.4f}")
print(f"F-stat  = {F_stat:.2f}")
print(f"p-value = {p_val:.2e}")
```

```
R²      = 0.5722
F-stat  = 65.09
p-value = 8.88e-27
```

Model Selection

Model Selection

What is model selection?

! Model selection

Model selection is the application of a principled method to determine the complexity of the model, e.g. choosing a subset of predictors, choosing the degree of the polynomial model etc.

- A strong motivation for performing model selection is to avoid overfitting!

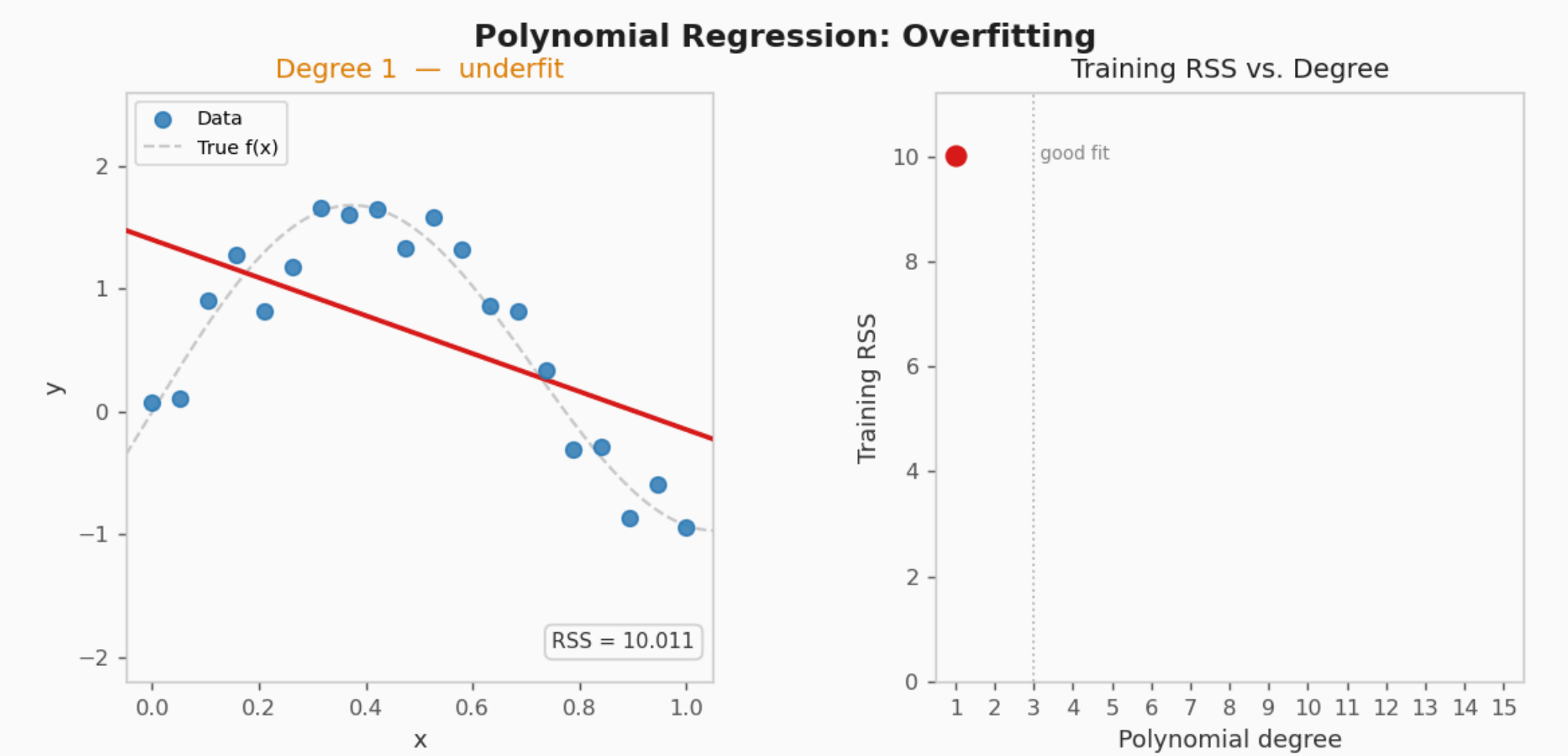
Overfitting

! Overfitting

Overfitting is the phenomenon where the model is **unnecessarily complex**, in the sense that portions of the model captures the random noise in the observation, rather than the relationship between predictor(s) and response.

- Overfitting causes the model to lose predictive power on new data.

Overfitting Visualization



Overfitting visualization.

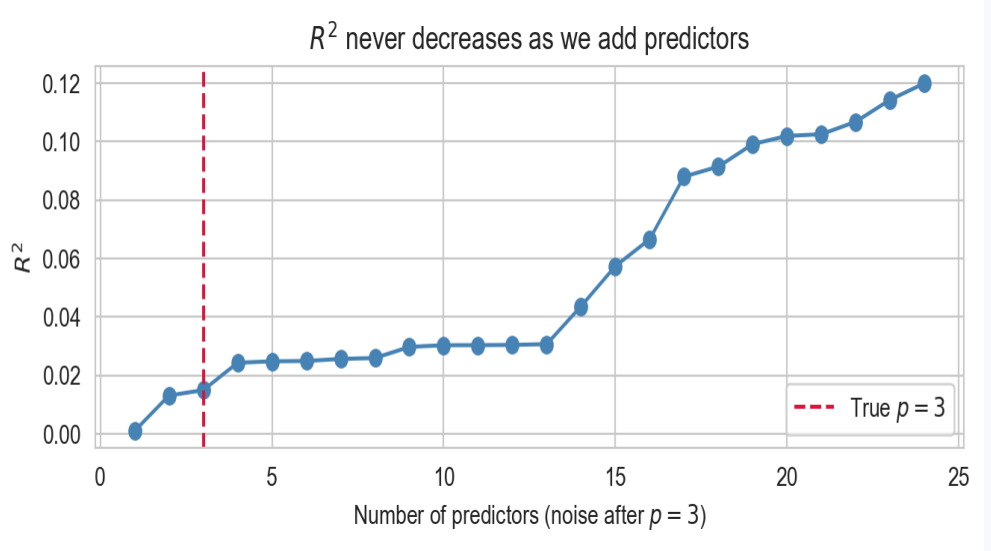
- We will consider model selection in a later lecture but we will briefly discuss some common model selection criteria in this lecture.

Why R^2 Is Not Enough in MLR

R^2 Always Increases

Adding a predictor to a model **never decreases** R^2, even if the predictor is pure noise.

```python
rng3 = np.random.default_rng(99)
R2_vals = []
Xk_full = rng3.normal(0, 1, (n, 25))
for k in range(1, 25):
    Xk = np.column_stack([
      np.ones(n),
      Xk_full[:, :k]
    ])
    bk = np.linalg.solve(Xk.T @ Xk, Xk.T @ y)
    yk = Xk @ bk
    R2_vals.append(1 - np.sum((y - yk)**2) / SST)

plt.figure(figsize=(7, 3))
plt.plot(range(1, 25), R2_vals, 'o-', color='steelblue')
plt.axvline(3, color='crimson', lw=1.5, linestyle='--',
            label='True $p = 3$')
plt.xlabel('Number of predictors (noise after $p=3$)')
plt.ylabel('$R^2$')
plt.title('$R^2$ never decreases as we add predictors')
plt.legend(); plt.tight_layout(); plt.show()
```



Adjusted R^2

Adjusted R^2

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n-p-1)}{\text{SST}/(n-1)} = 1 - (1-R^2)\frac{n-1}{n-p-1}.$$

- Penalises for adding predictors: compares **mean squared errors**, not raw sums.
- $\bar{R}^2$ **can decrease** when an added predictor does not reduce SSE enough to offset the lost degree of freedom.
- Use $\bar{R}^2$ (not R^2) to compare models with different numbers of predictors.

AIC and BIC

Information criteria balance goodness of fit against model complexity.

$$\text{AIC} = n \log\left(\frac{\text{RSS}}{n}\right) + 2(p+2), \qquad \text{BIC} = n \log\left(\frac{\text{RSS}}{n}\right) + (p+2)\log(n).$$

Criterion	Penalty term	Best used for
AIC	$2(p+2)$ — mild	Maximising predictive accuracy
BIC	$(p+2)\log n$ — stronger	Identifying the true model

- **Lower** is better.
- BIC penalises extra parameters more heavily for large n, so it selects sparser models.
- Unlike $\bar{R}^2$, AIC/BIC can compare non-nested models.

Partial F-Tests: Comparing Nested Models

A **partial F-test** compares a **full** model to a **reduced** (nested) model:

$$F = \frac{(\text{SSE}_{\text{red}} - \text{SSE}_{\text{full}})/q}{\text{SSE}_{\text{full}}/(n - p_{\text{full}} - 1)} \sim F_{q,\, n-p_{\text{full}}-1},$$

where q is the number of additional predictors in the full model.

When to use a partial F-test

- Test whether a **group** of predictors (e.g. all species dummies, all interaction terms) collectively improves fit.
- One individual t-test cannot capture the joint contribution of several correlated terms.
- In Python/statsmodels: `anova_lm(reduced_model, full_model)` — more on this in Lecture 12.

Multicollinearity

⚠ What Is Multicollinearity?

Multicollinearity occurs when two or more predictors are highly linearly correlated with each other.

What goes wrong?

- $\mathbf{X}^\top \mathbf{X}$ becomes **nearly singular** — its inverse has very large entries.
- This **inflates** standard errors $\widehat{\mathrm{SE}}(\hat{\beta}_j)$.
- Individual t-statistics become small (high p-values) even when predictors collectively predict Y well.
- The **sign** of $\hat{\beta}_j$ can be unstable across samples.

Important note

- Multicollinearity does **not** bias $\hat{\boldsymbol{\beta}}$ — estimates remain unbiased.
- It inflates **variance** — estimates become imprecise and unreliable for interpretation.
- The overall F-test may remain highly significant even when all t-tests are not.

Summary: SLR vs. MLR

Component	SLR ($p = 1$)	MLR (p predictors)
Model	$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$	$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$
OLS estimator	$\hat{\beta}_1 = S_{xy}/S_{xx}$	$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$
Fitted values	$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$	$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$
Var of $\hat{\beta}$	σ^2/S_{xx}	$\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$
Residual df	$n-2$	$n-p-1$
Inference	t_{n-2}	t_{n-p-1}; $F_{p,\,n-p-1}$
Fit metric	R^2	Adj. R^2, AIC, BIC

Conclusion

✅ What we covered

- **MLR model**: extending SLR to p predictors; partial slope interpretation (ceteris paribus).
- **Matrix formulation**: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$; the design matrix and Gauss-Markov in matrix form.
- **OLS derivation**: $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}$ from the normal equations; the hat matrix $\mathbf{H}$.
- **Properties**: unbiasedness, variance-covariance matrix $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, Gauss-Markov (BLUE).
- **Inference**: t-tests for individual slopes; overall F-test; partial F-tests for nested models.
- **Goodness of fit**: why R^2 is insufficient in MLR; adjusted R^2, AIC, BIC.
- **Multicollinearity**: causes, consequences, and detection via VIF.

📅 What's next?

References